

Labwork 2 Report: Get to Know Your GPU

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1 Introduction

In this labwork, I used the `numba.cuda` Python library to query and inspect the hardware specifications of the GPU available on my local machine.

2 Implementation

I created a Python script named `labwork2.py`. The code detects the CUDA-capable GPU, selects the primary device, and retrieves hardware details such as device name, multiprocessor count, and memory status.

Below is the Python code used: [language=Python] import numba from numba import cuda

```
def main(): if not cuda.is_available(): print("CUDA is not available") return
cuda.detect() device = cuda.select_device(0)
dev_id = device.id dev_name = device.name.decode('utf-8') if isinstance(device.name, bytes):
print("Device ID:", dev_id) print("Device Name: ", dev_name)
try: mp_count = device.MULTIPROCESSOR_COUNT print("Multiprocessors :
", mp_count) except AttributeError: pass
free_mem, total_mem = cuda.current_context().get_memory_info() print("Total Memory (bytes) :
", total_mem) print("Free Memory (bytes) : ", free_mem) print("Total Memory (GB) :
", total_mem/(1024**3)) print("Free Memory (GB) : ", free_mem/(1024**3))
if __name__ == "__main__": main()
```

3 Results

Running `labwork2.py` on my system produced the following output:

- **Device ID:** 0
- **Device Name:** NVIDIA GeForce GTX 1050
- **Compute Capability:** 6.1
- **Multiprocessors (SM):** 6
- **Total Memory:** 2.94 GB (3,158,114,304 bytes)
- **Free Memory:** 2.89 GB (3,100,508,160 bytes)
- **FP16 Support:** Yes

4 Conclusion

The script ran successfully on the local environment. The system's GPU is an NVIDIA GeForce GTX 1050 featuring 6 Streaming Multiprocessors and approximately 2.94 GB of VRAM.